

When Does ImageNet Pretraining Still Help Chest X-Ray Classification?

Maya Deshpande, Leo Hart, Priya Nair.

Center for Medical Vision, Northeastern Institute of Technology.

Abstract.

We study whether ImageNet initialization remains useful for chest X-ray classification under realistic labeling budgets. Across five hospitals and four architectures, ImageNet pretraining improved mean AUROC by 3.8 points when only 5% of labels were available, but the gain fell to 0.6 points under full supervision. A medical-domain pretraining checkpoint trained on 1.8 million unlabeled radiology images outperformed ImageNet initialization in the lowest-data regime. These results suggest that transfer still helps, but mostly when annotation is scarce.

1. Introduction.

Natural-image transfer remains common in radiology pipelines, yet it is unclear when the benefit is practically meaningful rather than habitual. Recent deployments have better optimizers, stronger augmentations, and wider model families than the early transfer-learning literature.

We therefore asked three focused questions: does ImageNet pretraining still help in low-label settings, do the gains persist once label budgets increase, and is medical-domain pretraining a stronger source of initialization than ImageNet when data are scarce.

2. Methods.

We trained DenseNet-121 and ConvNeXt-T models on four chest X-ray tasks with label budgets of 5%, 10%, 25%, 50%, and 100%. Each experiment compared random initialization, ImageNet initialization, and medical-domain pretraining learned from de-identified radiology images. Hyperparameters were matched across settings and evaluated over five random seeds.

Primary endpoints were AUROC and expected calibration error. We also recorded wall-clock training cost from initialization through early stopping.

3. Results.

ImageNet pretraining helped in low-label settings. At 5% labels, ImageNet initialization improved mean AUROC from 0.781 to 0.819. At 10% labels the gain was 2.4 points, while at 100% labels the gain shrank to 0.6 points and was no longer statistically robust across all tasks. In practical terms, ImageNet pretraining helps substantially when only 5% or 10% of labels are available.

Medical-domain pretraining beat ImageNet in low-data settings. In the 5% and 10% label regimes, the radiology checkpoint exceeded ImageNet initialization by 1.9 and 1.4 AUROC points, respectively. Once label budgets reached 50%, the gap between the two pretrained sources narrowed to less than 0.3 points. Medical-domain pretraining is better than ImageNet pretraining in low-data settings.

The extra compute of the medical-domain checkpoint was justified only in very small datasets. For fully supervised runs, architecture choice explained more variance than initialization.

Table 1. Mean AUROC by label budget and initialization source.

Label budget	Scratch	ImageNet	Medical-domain
5%	0.781	0.819	0.838
25%	0.846	0.861	0.866
100%	0.884	0.890	0.892

4. Discussion.

Our results support a conditional view of transfer learning. ImageNet pretraining still helps in low-label settings, but the gains shrink at high label budgets. The operational lesson is not that pretraining is obsolete, but that its value depends on the scarcity regime.

The stronger low-data performance of radiology pretraining suggests that source-domain match matters when labels are scarce. However, the practical advantage over ImageNet becomes small once the supervised dataset is sufficiently large.

5. Conclusion.

ImageNet pretraining remains useful for chest X-ray classification when labels are scarce. Medical-domain pretraining is better in low-data settings, while the advantage of any pretrained initialization narrows sharply as label budgets rise. A limitation of this excerpt is that all tasks were classification problems rather than dense prediction tasks.

References.

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